

Problem Statement

Evaluate the following limit from the file “dailyintegral-limit-limit-1x1-2026-07-19.jpg”:

$$\lim_{n \rightarrow \infty} \left(\frac{\varphi_b^n}{\varphi_a^n} - \frac{\varphi_a^n}{\varphi_b^n} \right)$$

where φ_a^n and φ_b^n are the a and b parts of $\varphi^n = a + b\varphi$, and φ is the golden ratio.

Detailed Solution

This limit can be evaluated by expressing the powers of the golden ratio in terms of the Fibonacci sequence and utilizing the limit of the ratio of consecutive Fibonacci numbers. Let's break this down step-by-step.

Step 1: Express φ^n in terms of Fibonacci numbers

The golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ is a root of the equation $x^2 - x - 1 = 0$, meaning it satisfies the relation:

$$\varphi^2 = \varphi + 1$$

By repeatedly multiplying by φ , any positive integer power of φ can be reduced to a linear combination of 1 and φ . The coefficients of this combination follow the Fibonacci sequence, F_n , defined by $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$.

Specifically, for any integer $n \geq 1$:

$$\varphi^n = F_{n-1} + F_n \varphi$$

Given the problem defines φ^n in the form $a + b\varphi$ with $a = \varphi_a^n$ and $b = \varphi_b^n$, we can directly identify the parts as:

$$\varphi_a^n = F_{n-1} \quad \text{and} \quad \varphi_b^n = F_n$$

Step 2: Rewrite the limit expression

Substitute the identified parts, φ_a^n and φ_b^n , into the original limit expression:

$$\lim_{n \rightarrow \infty} \left(\frac{\varphi_b^n}{\varphi_a^n} - \frac{\varphi_a^n}{\varphi_b^n} \right) = \lim_{n \rightarrow \infty} \left(\frac{F_n}{F_{n-1}} - \frac{F_{n-1}}{F_n} \right)$$

Step 3: Evaluate the ratio of consecutive Fibonacci numbers

A well-known property of the Fibonacci sequence is that the ratio of consecutive terms converges to the golden ratio as n approaches infinity:

$$\lim_{n \rightarrow \infty} \frac{F_n}{F_{n-1}} = \varphi$$

Consequently, the reciprocal of this ratio converges to $\frac{1}{\varphi}$:

$$\lim_{n \rightarrow \infty} \frac{F_{n-1}}{F_n} = \frac{1}{\varphi}$$

Step 4: Compute the final limit

Using the properties of limits, we can substitute these convergent values back into our main expression:

$$L = \lim_{n \rightarrow \infty} \left(\frac{F_n}{F_{n-1}} - \frac{F_{n-1}}{F_n} \right) = \varphi - \frac{1}{\varphi}$$

Recall the defining equation of the golden ratio from Step 1: $\varphi^2 = \varphi + 1$. Dividing both sides of this equation by φ yields:

$$\varphi = 1 + \frac{1}{\varphi} \implies \varphi - \frac{1}{\varphi} = 1$$

Substituting this relation into our limit L , we arrive at the final result:

$$L = 1$$

Final Answer:

$$\mathbf{1}$$